

Assignment 8 - C++ - classes

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15.12.2023

Poengsum frakoblet modus:

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Anonym vurdering: **nei**

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▼ Detaljer

Introduction

Note: This task is an advanced programming task that should only be attempted after you are satisfied with your results on the other tasks.

This task combines pointers with C++ classes and thus requires an understanding of both.

8.1

Directory name: ikt101g23h/assignments/solutions/assignment_8_1

Relevant lectures: Pointers, C++ classes

Description

Implement a linked list that stores integers.

Start by reading and understanding the general description of a linked list:

https://en.wikipedia.org/wiki/Linked_list  (https://en.wikipedia.org/wiki/Linked_list)

Create a class for the list and a class for the node. You must then figure out how to add a new node when the user calls `push_back()`, and how to remove a node when the user calls `remove()`.

Limitations

Each node is connected to the next node with pointers. The following are not allowed in this assignment:

- arrays (for the list itself, OK elsewhere)
- `std::vector`
- `std::list`
- `std::map`
- (any other `std::` container)

Specification

The list class must contain the following functions:

- `void push_back(const int value)`
Add a value to the back of your list
- `size_t size()`
Get the item count
- `int& at(size_t index)`
Retrieve item by index
- `void remove(size_t index)`
Remove item at specified index

Areas of interest: Pointers, classes, new/delete operators

Application UI

Once you have implemented your linked list write an application that uses the list with the following menu.

The user types a number and presses enter to choose:

1. Add integer (asks for a number and adds it to the list)
2. Show integer at index (ask for index)
3. Remove integer at index (ask for index)
4. Print information (see below)
5. Exit

Note: Indexes start from 0.

Expected output with no integers:

List contains no integers

Expected output (from 2):

Integer at index 5: 42

Expected output (from 4):

List contains 4 integers: 1 2 3 4

Test locally

see *Local testing with Docker* under Assignments.

Test the solution on your own computer. You can fix your code and retest as many times as you want.

Deliver and test in Bamboo

See *Delivery and Bamboo testing* under Assignments.

Finally deliver and test the solution in Bamboo. This is what updates your score in Canvas which is the score I can see. You can test as many times as you want here as well up until the final deadline.